

# Stacks & Queues — Short Revision Notes (DS)

Data Structures (DS) · Unit 2 · Stacks & Queues · 2025

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## Overview

These notes cover **Stacks & Queues**. Core data structures and the algorithms built on them — arrays, lists, stacks, trees, graphs, sorting and hashing. Each section states the core idea, a worked intuition and the points examiners look for.

### 1. Stack ADT

A stack is a LIFO structure supporting push, pop and peek in constant time.

**Why it matters:** Stack ADT appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of stack adt in one line.
- A small example showing stack adt in practice.
- Common mistakes students make and how to avoid them.

### 2. Infix to Postfix Conversion

Operator precedence and a stack convert infix expressions to postfix for easy evaluation.

**Why it matters:** Infix to Postfix Conversion appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of infix to postfix conversion in one line.
- A small example showing infix to postfix conversion in practice.
- Common mistakes students make and how to avoid them.

### 3. Recursion and the Call Stack

Each recursive call pushes a frame; tail recursion can be converted to iteration.

**Why it matters:** Recursion and the Call Stack appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of recursion and the call stack in one line.
- A small example showing recursion and the call stack in practice.
- Common mistakes students make and how to avoid them.

## 4. Queue ADT

A queue is FIFO; circular arrays avoid wasting space after dequeues.

**Why it matters:** Queue ADT appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of queue adt in one line.
- A small example showing queue adt in practice.
- Common mistakes students make and how to avoid them.

## 5. Deque and Priority Queue

Dequeues allow insertion and deletion at both ends; priority queues serve the highest-priority element first.

**Why it matters:** Deque and Priority Queue appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of deque and priority queue in one line.
- A small example showing deque and priority queue in practice.
- Common mistakes students make and how to avoid them.

## 6. Applications of Stacks and Queues

Balanced parentheses, undo operations, BFS and CPU scheduling all rely on stacks or queues.

**Why it matters:** Applications of Stacks and Queues appears regularly in mid and semester examinations. Be ready to define it precisely, draw a diagram where relevant, and give one realistic example.

- Key definition of applications of stacks and queues in one line.
- A small example showing applications of stacks and queues in practice.
- Common mistakes students make and how to avoid them.

## Quick revision

Stack ADT · Infix to Postfix Conversion · Recursion and the Call Stack · Queue ADT · Deque and Priority Queue · Applications of Stacks and Queues